Interest Rate, Bond Duration, and Financial Stability

How Federal Funds Rate Changes Influence
Bond and TIPS Returns

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Motivation Policy Questions

How do changes in the Federal Funds Rate influence bond market returns?

Subquestions:

- Which bonds are most sensitive to interest rate changes?
- Does bond duration amplify interest rate risk?
- Do inflation-protected securities (TIPS) behave differently?

- Monetary policy influences trillions of dollars in financial assets
- Rapid rate increases can create market instability
- Understanding interest-rate sensitivity helps monitor financial risk
- Duration exposure is a key risk factor in bond markets

Datasets

Dataset 1:

WRDS Bond Returns

Key Variables:

- Bond prices and returns
- Duration and maturity
- Yield and coupon information
- Credit ratings
- Issuer identifiers

Dataset 2:

Federal Reserve interest rate dataset

Key variables:

- Federal Funds Rate
- Treasury yield curve
- Monthly interest rate changes

Dataset 3:

CRSP Treasury securities dataset

Key Variables:

- Treasury bond yields
- Duration and maturity
- Returns for Treasury and TIPS securities

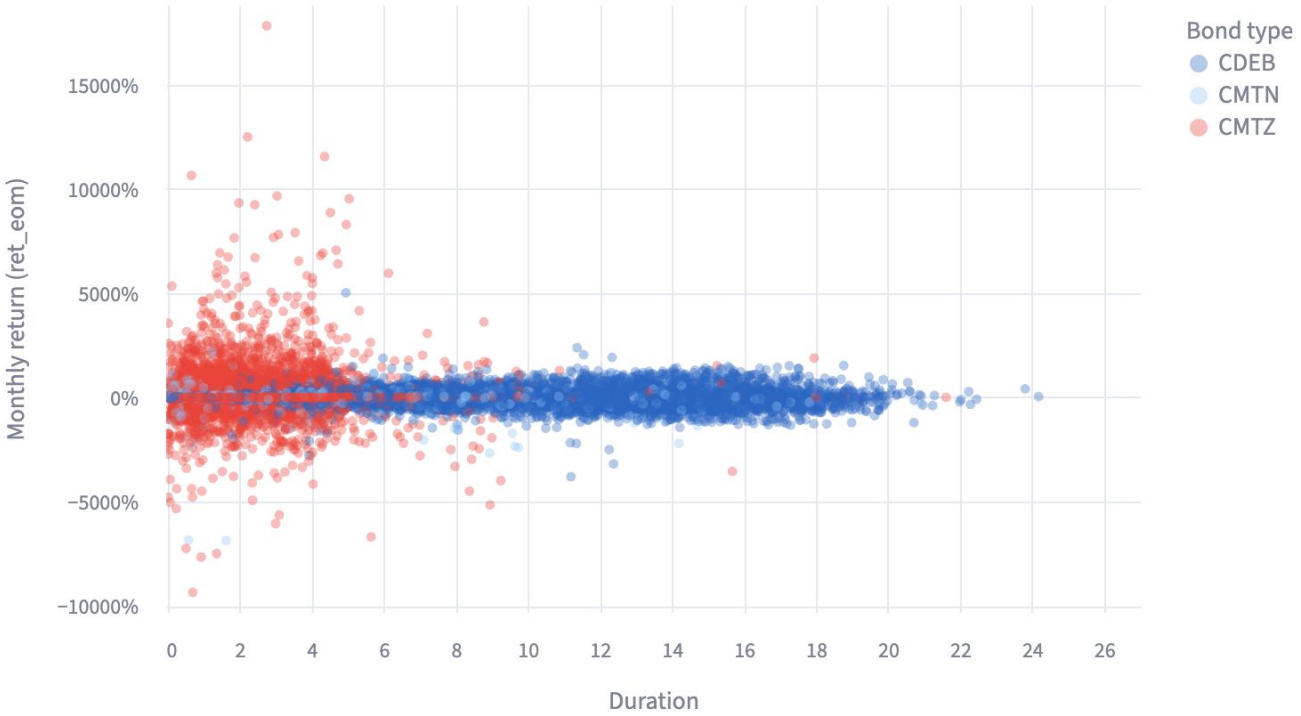
Methodology

Theory: Bond price sensitivity ≈ Duration × Rate change

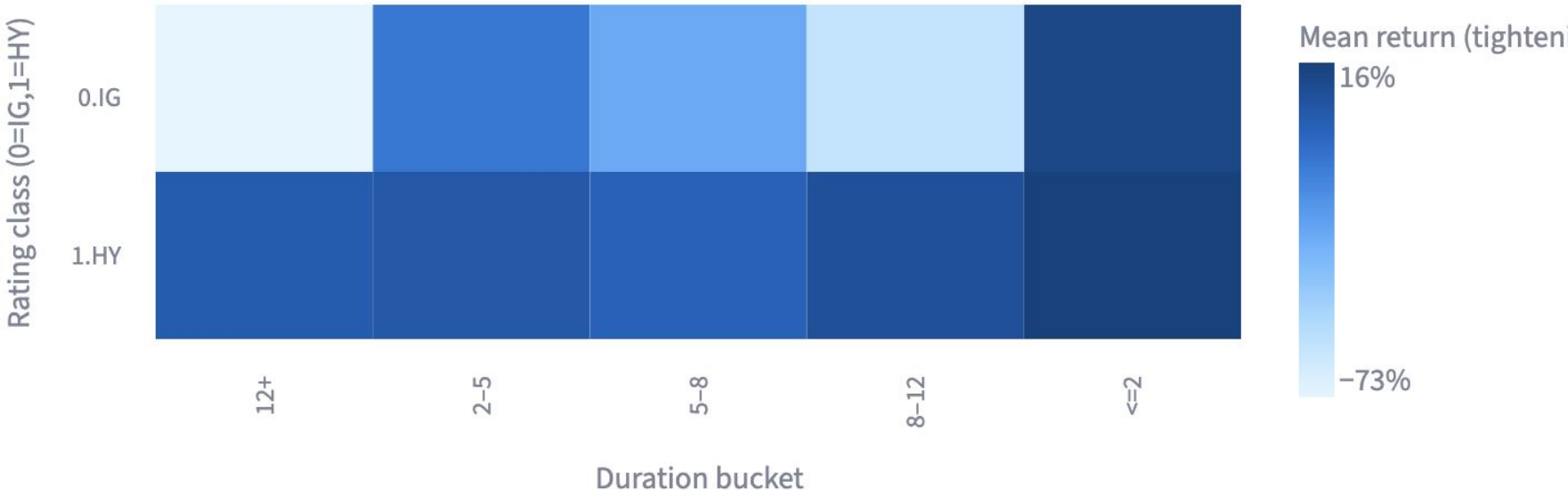
Data Processing:

1. Merge bond-level data with interest rate data
2. Compute monthly changes in the Federal Funds Rate
3. Group bonds by duration and maturity
4. Analyze how returns respond to interest rate changes

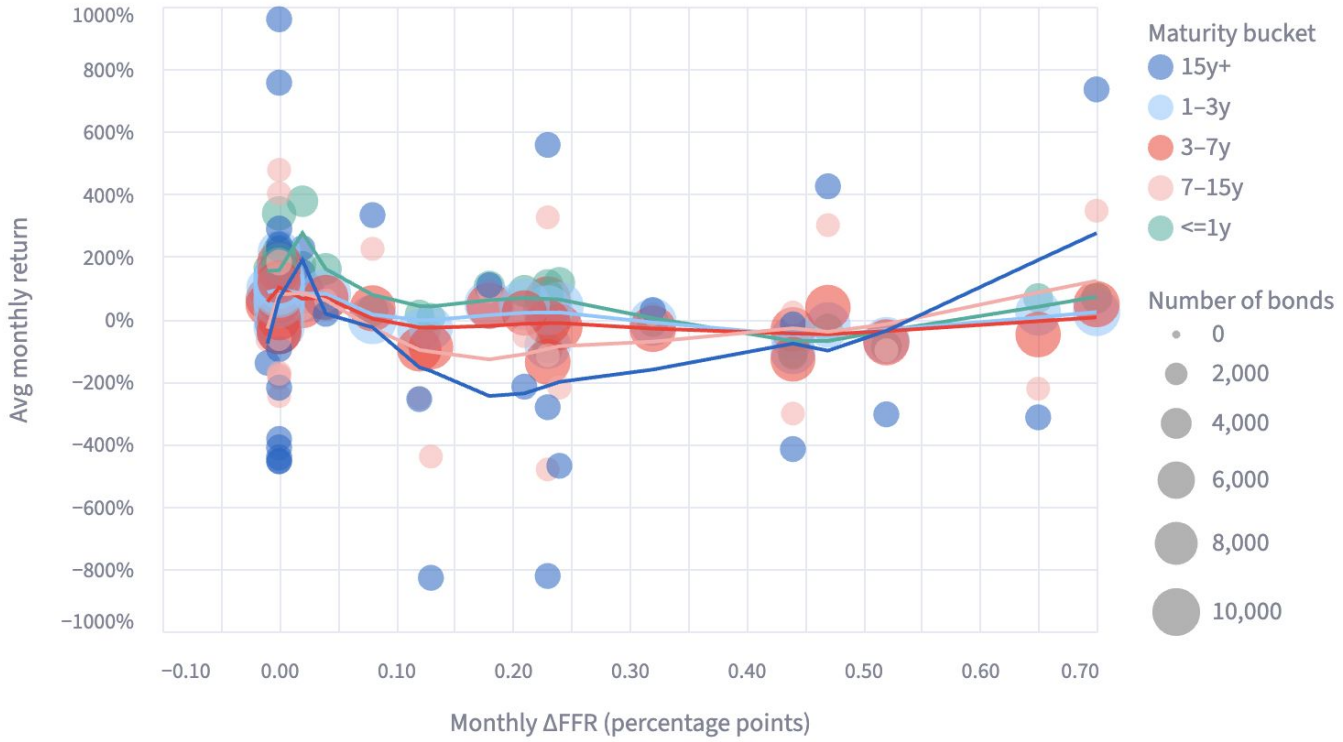
Visualization 1: Duration VS Bond Reture



Visualization 2: Tightening Heatmap



Visualization 3: Federal Funds VS Returns



Findings and Policy Implications

Findings:

- Longer duration bonds show greater sensitivity to interest rate changes
- Returns become more volatile during tightening periods
- TIPS may respond differently due to inflation protection
- Longer maturity bonds show larger reactions to policy shocks

Policy Implications

- Rapid monetary tightening can create stress in bond markets
- Duration risk amplifies market volatility
- Monitoring duration exposure helps assess financial stability risks

Dashboards

https://yuxiliang1727-final-project-w26-app-wigugz.streamlit.app/